Week 3 Presentation Content

Machine learning, evaluation and dashboard draft

Presentation goal

Present functioning and evaluated K Means and Logistic Regression models, persist their results in MySQL and show an initial decision dashboard.

Deck map

Slide	Purpose
1	Weekly status and time review
2	Experimental design and leakage controls
3	K Means workflow
4	K selection and segment profiles
5	Logistic Regression workflow
6	Classification results
7	MySQL writeback
8	Initial dashboard
9	Paper progress and Week 4 plan

Evidence to keep open during Q and A

- kmeans/3.csv through kmeans/7.csv and tmp/report/figures/clustering_summary.png
- Model metrics and confusion matrix in the repository result files
- MySQL tables user_segments, rating_predictions and model_metrics

How to use this file Copy the on-slide text into PowerPoint. Use the visual direction to build the graphic, and rehearse from the speaker cue. Update blank actual-effort fields before presenting.

Slide 1 Week 3 turns governed data into evaluated evidence

On-slide copy

- WHAT: engineer features, evaluate K Means and Logistic Regression, write results to MySQL and connect the dashboard.
- WHEN: freeze model outputs before Week 4 recommendations and final reporting.
- WHY: decisions need out-of-time performance, segment interpretation and a traceable writeback contract.
- PLANNED EFFORT: 20-24 team hours. ACTUAL EFFORT: _____ hours. GAP: _____ hours.
- ACTION FROM GAP: _____

Visual direction Use a four-stage strip: features -> models -> evaluation -> dashboard.

Speaker cue Start with the experimental contract, then show results. This prevents the metrics from appearing detached from methodology.

Slide 2 Chronological splits and TRAIN-only features limit leakage

On-slide copy

- TRAIN: 7,000,042 ratings from 9 January 1995 to 2 April 2005.
- VALID: 1,500,006 ratings from 2 April 2005 to 28 December 2006.
- TEST: 1,500,006 ratings from 28 December 2006 to 5 January 2009.
- TRAIN rows use leave-one-out user and movie aggregates; VALID and TEST use frozen TRAIN aggregates.
- user_id and movie_id are identifiers and are not Logistic Regression predictors.

Visual direction Use a time axis with hard split boundaries and a TRAIN-only feature arrow.

Speaker cue Explain the rule in one sentence: no row may use information that would not have existed at its prediction time.

Slide 3 K Means segments warm-start users

On-slide copy

- Create user-level TRAIN features for activity, average rating and genre affinity.
- Standardize numeric features so high-volume fields do not dominate Euclidean distance.
- Compare K values from 3 through 7 using mean silhouette and cluster interpretability.
- Assign business labels only after inspecting final centroids and behavior profiles.
- Persist model version, cluster number and final segment label for each eligible user.

Visual direction Show the KNIME pipeline beside a small feature table. Avoid listing every node.

Speaker cue Clarify that segment names summarize observed profiles; they are not demographic identities.

Slide 4 K equals 4 balances separation and interpretability

On-slide copy

- Mean silhouette: K=3 0.06021; K=4 0.06864; K=5 0.06919; K=6 0.06460; K=7 0.06511.
- K=5 improves mean silhouette by only 0.00055 over K=4, while K=4 gives clearer decision profiles.
- 54,873 users are segmented, equal to 78.5% of rating users.
- Genre Specialists are the largest segment at 17,346 users and have a 71.43% positive-rating rate.
- Selective Critics are the most active, averaging 385.3 TRAIN ratings per user.

Segment	Users	Positive rate	Profile cue
Diverse Explorers	12,182	51.38%	Thriller and action affinity
Selective Critics	10,888	50.88%	Highest activity
Mainstream Enthusiasts	14,457	42.06%	Family and animation affinity
Genre Specialists	17,346	71.43%	Largest and most positive

Visual direction Use tmp/report/figures/clustering_summary.png. Highlight the selected K=4 point and two segment facts.

Speaker cue A low silhouette means the segments overlap. Use them as decision lenses, not as perfectly separated natural groups.

Slide 5 Logistic Regression predicts positive engagement

On-slide copy

- Target: 1 when rating >= 4.0; otherwise 0.
- Predictors use prior user behavior, prior movie behavior, activity and context derived from TRAIN.
- TRAIN leave-one-out aggregation prevents each rating from leaking into its own predictors.
- VALID selects settings and TEST supplies the final reported metrics.
- Prediction rows store probability, class, actual outcome, split and model version.

Visual direction Use a feature-to-probability pipeline ending in a 0.5 decision threshold.

Speaker cue Logistic Regression was chosen because coefficients and probabilities are easier to explain and govern than a black-box alternative.

Slide 6 The model beats the majority-class baseline on warm-start TEST rows

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- Accuracy 69.26% versus 59.32% baseline, an improvement of 9.93 percentage points.
- AUC 75.83%, precision 60.58%, recall 69.94% and F1 64.92%.
- Confusion matrix: TN 31,273; FP 14,189; FN 9,369; TP 21,802.
- TEST feature coverage is 76,633 of 1,500,006 rows, or 5.11%; 94.89% are excluded by cold-start rules.
- The performance claim therefore applies only to covered warm-start rows.

Visual direction Use tmp/report/figures/model_evaluation.png and place coverage beside the metrics, not in a footnote.

Speaker cue Lead with baseline improvement, then immediately qualify it with coverage. This is the central governance message.

Slide 7 Model outputs return to MySQL with version fields

On-slide copy

- user_segments stores user_id, cluster, segment label and model version.
- rating_predictions stores probability, predicted class, actual class, split and version.
- model_metrics stores metric name, metric value, evaluation split and run metadata.
- Dashboard views join stable dimensions to these result tables without recalculating the models.
- The writeback step creates one auditable handoff from analytics to reporting.

Visual direction Show three result tables feeding one dashboard-view layer.

Speaker cue Demonstrate row counts and one sample record from each output table rather than scrolling through raw results.

Slide 8 The dashboard draft follows five decisions

On-slide copy

- Executive Overview: dataset scale, positive rate, model status and key actions.
- Audience Segments: segment size, behavior and genre affinity.
- Positive Engagement Model: baseline comparison, metrics, confusion matrix and coverage.
- Content Opportunity Matrix: predicted appeal, actual response, popularity and support.
- Model Governance: versions, split logic, thresholds, coverage and limitations.
- Global controls: model version, segment, genre and minimum support.

Visual direction Use tmp/report/figures/dashboard_map.png as a five-page navigation map.

Speaker cue Explain what decision each page supports. Avoid describing visuals without linking them to an action.

Slide 9 Week 4 will convert model evidence into bounded recommendations

On-slide copy

- Paper progress: methodology, experimental design and preliminary results are ready for the IEEE draft.
- Finalize Power BI interactions and validate that every visual uses versioned MySQL views.
- Create at least three recommendations with evidence, owner, test design, assumptions and risks.
- Complete repository instructions, final report, live demo path and Q and A backup slides.
- Carry-forward issue or decision: _____

Effort item	Planned	Actual	Gap and action
Feature and model work	_____ h	_____ h	_____
Writeback and dashboard	_____ h	_____ h	_____
Paper and rehearsal	_____ h	_____ h	_____

Visual direction Use a Week 4 acceptance checklist and the effort table below.

Speaker cue End by agreeing on the freeze point for data, model version and dashboard refresh.